INCH-POUND
MIL-PRF-32494
18 March 2014

PERFORMANCE SPECIFICATION

LIGHT UNIT, PORTABLE, TF-2 Lightcart

This specification is approved for use by the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.
This specification is for the design, construction, and delivery of a Light Unit, Portable; herein, refer to as a TF-2 light cart. The TF-2 light cart is primarily designed for field (outside) use at forward operating military bases.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-595/33446	Yellow, Flat or Lusterless
FED-STD-595/24052	Dark Green, Semigloss

Comments, suggestions, or questions on this document should be addressed to: AFLCMC/WNZEC, 235 BYRON ST STE 19A, Robins AFB GA 31098-1813. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .

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FED-STD-595/26173	Gray, Semigloss
FED-STD-595/37038	Miscellaneous, Flat or Lusterless
FED-STD-595/31136	Red, Flat or Lusterless

COMMERCIAL ITEM DESCRIPTIONS (CID)

A-A-52464	Coupler, Drawbar, Ring; Light Duty, 60,000 lb GVW; Offset (Taper Shank), 60,000 lb GVW; and Heavy Duty 120,000 lb GVW
A-A-52550	Pintle Assembly, Towing, Manual Release, 18,000, 40,000, and 100,000 lbs Capacity
A-A-52557	Fuel Oil, Diesel; for Posts, Camps, and Stations

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-B-18013	Battery, Storage, Support Equipment
MIL-DTL-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-DTL-25959	Tie Down, Tensioners, Cargo, Aircraft
MIL-DTL-53030	Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free
MIL-DTL-81706	Chemical Conversion Materials for Coating Aluminum and Aluminum Alloys
MIL-DTL-83133	Turbine Fuels, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)
MIL-PRF-2104	Lubricating Oil, Internal Combustion Engine, Combat/ Tactical Service
MIL-PRF-26915	Primer Coating, For Steel Surfaces
MIL-PRF-23377	Primer Coatings, Epoxy, High-Solids
MIL-PRF-27260	Tie Down, Cargo, Aircraft, CGU-1/B
MIL-PRF-46167	Lubricating Oil, Internal Combustion Engine, Arctic
MIL-PRF-85285	Coating: Polyurethane, Aircraft and Support Equipment

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-209	Lifting and Tiedown Provisions
MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-705	Generator Sets, Engine Driven Methods of Tests and Instructions
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-882	System Safety
MIL-STD-889	Dissimilar Metals
MIL-STD-1366	Interface Standard for Transportability Criteria
MIL-STD-1472	Human Engineering

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MIL-STD-1474	Noise Limits
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-781	Handbook for Reliability Test Methods, Plans, and Environments for Engineering, Development, Qualification, and Production

(Copies of these documents are available online at http://quicksearch.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR FORCE PUBLICATIONS

AFI 32-7086	Hazardous Materials Management

(Copies of the publications are available online at http://www.e-publishing.af.mil/, while the drawing is available from the procuring activity.)

TEST OPERATIONS PROCEDURE (TOP)

TOP 2-2-800	Center of Gravity

(Copies of this document are available online at https://vdls.atc.army.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL (ASTM)

ASTM D975	Standard Specification for Diesel Fuel Oil
ASTM D1655	Standard Specification for Aviation Turbine Fuels

(Application for copies should be addressed to ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken PA 19428-2959, www.astm.org.)

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AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	Structural Welding Code–Steel
AWS D1.2/D1.2M	Structural Welding Code–Aluminum
AWS D1.3/D1.3M	Structural Welding Code–Sheet Steel
AWS D1.6/D1.6M	Structural Welding Code–Stainless Steel

(Application for copies should be addressed to American Welding Society, 550 N.W. LeJeune Road, Miami FL 33126, www.aws.org)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA-250	Enclosures for Electrical Equipment (1000V Maximum)
NEMA-MG1	Motors and Diesel engines
NEMA-WD1	General Requirements for Wiring Devices
NEMA-WD6	Wiring Devices-Dimensional Specifications

(Application for copies should be addressed to NEMA, 1300 North 17th Street, Suite 1847, Rosslyn VA 22209, www.nema.org.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	National Electrical Code

(Application for copies should be addressed to NFPA, 1 Batterymarch Park, Quincy MA 02169-7471, www.nfpa.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE-ARP1247	General Requirements for Aerospace Ground Support Equipment, Motorized and Non-motorized
SAE-J1292	Automobile, Truck, Truck-Trailer, Trailer, and Motor Coach Wiring
SAE-AS8090	Towed Aerospace Ground Equipment, General Specification For

(Application for copies should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale PA 15096, www.sae.org)

TIRE AND RIM ASSOCIATION

TRA-YB	Tire and Rim Association, Year Book

(Application for copies should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Ave., Suite 150, Copley OH 44321, www.us-tra.org)

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UNDERWRITERS LABORATORIES, Inc. (UL)

UL486A	Wire Connectors and Soldering Lugs for Use with Copper Conductors
UL498	UL Standard for Safety Attachment Plugs and Receptacles
UL1077	UL Standard for Safety Supplementary Protectors for Use in Electrical Equipment
UL489	Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures

(Application for copies should be addressed to Underwriters Laboratories Inc. (UL), 333 Pfingsten Road, Northbrook IL 60062-2096, www.ul.com)

FOREIGN SPECIFICATIONS

GOST 10227-86	Fuels for Jet Engines

(A copy of GOST 10227-86 can be obtained from the Procuring Contracting Officer (PCO).)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence, with the exception of the appended engine specification. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the solicitation, (see 6.2), first article light cart(s) shall be subjected to first article inspection in accordance with 4.2.

3.2 Light cart description. The light cart is designed to provide remote area lighting capability at forward air bases and fixed air bases worldwide under a variety of climatic conditions with minimal logistics support. The light cart is self-contained, towable light source consisting of a durable metal enclosure mounted on a two-wheel chassis that shall meet the requirements of SAE AS8090 Type III, Group A vehicles. The diesel engine provides power to four 1000 watt metal-halide lights and auxiliary power outlets as described in the following paragraphs. The four light fixtures are mounted on a mechanical telescoping tower that can be raised from 15 feet to 30 feet by a single operator. When fully retracted, the tower is hinged to permit lowering to the horizontal position for transportation.

3.3 Design and construction. The design shall promote cost effective, life-cycle sustainability by addressing considerations such as incorporating open standards, reducing pollutant emissions and wastes, and increasing fuel economy, while satisfying system performance requirements. The light cart shall be designed and constructed in accordance with all applicable European Union (EU) requirements in order to have the “CE” marking affixed (see 3.3.3.1). It shall be

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designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. It shall be weatherproof and designed to prevent the intrusion of water, sand, and dust into critical operating components.

3.3.1 Materials, protective coatings, and finish.

3.3.1.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials shall be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. Used, rebuilt, or refurbished items shall not be provided.

3.3.1.2 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the light cart shall not be used or protective coating shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. Protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used. Fasteners, handles, and fittings used in the assembly of the light cart shall also be primed and painted as specified in 3.3.1.2 and subordinate paragraphs. Commercial items, as defined in 6.3.1, and used as components of the light cart (but not the light cart itself) may be prepared and coated in accordance with the manufacturer’s standard practice, provided it is compatible with the exterior finish color and does not compromise operations in a contingency environment. See 3.3.4.2 for prohibited hazardous materials.

3.3.1.2.1 Surface preparation and pretreatment. Surface preparation and pretreatment shall be in accordance with the respective primer and topcoat specifications. Structures shall be cleaned, degreased, and scuffed or blasted prior to priming; primer shall be applied before any oxidation or rusting occurs. Aluminum surfaces shall have MIL-DTL-81706, Type II, Class 1A, chemical conversion coating applied in accordance with the manufacturer’s directions prior to priming.

3.3.1.2.2 Primer. Raw metal edges, to include fastener and drain holes, shall be coated with a MIL-DTL-53022 primer before applying topcoat. Surfaces shall have the appropriate primer for the surface composition type.

3.3.1.2.2.1 Ferrous surfaces. Ferrous structures and surfaces shall be primed with a water reducible zinc rich primer in accordance with MIL-PRF-26915, Type II, Class B; this shall be followed, within four hours, by a coat of MIL-DTL-53030 intermediate primer in a wet-to-wet primer application. This two part primer system shall yield a dry-film thickness of 2.0-2.5 mils for the zinc primer and 0.9 to 1.1 mils for the intermediate primer. The two-primer system shall be allowed to dry and fully cure in accordance with the primer manufacturer's directions prior to top coating.

3.3.1.2.2.2 Aluminum surfaces. Aluminum and mixed aluminum and ferrous structures and surfaces shall be primed with an epoxy primer, Type II, Class N of MIL-PRF-23377. This single part primer system shall yield a dry-film thickness of 0.6 to 0.8 mils.

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3.3.1.2.3 Topcoat. Topcoat shall be polyurethane in accordance with Type I, Class H of MIL-PRF-85285. Neither Chemical Agent Resistant Coating (CARC) nor powder coating shall be used. Topcoat shall be applied to a dry film thickness of 1.6 to 2.4 mils in all instances, regardless of the primer system utilized. The coating shall be free from runs, sags, orange peel, or other defects.

3.3.1.3 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be in contact with each other. Metal plating or metal spraying of dissimilar base metals to provide electromotively compatible abutting surfaces (for conductivity) is acceptable. The use of dissimilar metals when separated by suitable insulating material is permitted, except in systems where bridging of insulation materials by an electrically conductive fluid can occur. Sealants or gel type gasket materials shall be used between faying surfaces and butt joints.

3.3.1.4 Finish. Unless otherwise specified, the exterior finish and colors of the light cart shall be as defined in FED-STD-595: Semigloss Dark Green, color number 24052; Flat or Lusterless Tan 686A, color number 33446; or Semigloss Gray, color number 26173.

3.3.1.5 Exclusion of water. The design of the light cart shall be such as to prevent water leaking into, or being driven into, any part of the light cart interior when either in an operating or travelling configuration. All windows, doors, panels, covers, etc., shall be provided with sealing arrangements such that the entry of water is minimized when these items are correctly closed. Particular care shall be taken to prevent wetting of equipment and heat and sound proofing materials. Sharp corners and recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized corrosion attack. Sealed floors with suitable drainage shall be provided for storage compartments, engine compartments, and other areas in the light cart that could collect and retain water. Means shall be provided for drainage of water entrapped within the storage compartment resulting from operation with doors or covers open.

3.3.1.5.1 Fluid traps and faying surfaces. There shall be no fluid traps on the light cart. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

3.3.1.5.2 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

3.3.1.5.3 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to effect maximum drainage of accumulated fluids. The number and location of drain holes shall be sufficient to permit drainage of all fluids when the unit is in a five degrees (5°) incline in any plane. The minimum size of the drain holes shall be 0.25 inch.

3.3.2 Markings. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1-inch high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible, but shall not be less than ½-inch high. Markings, Information/Caution shall be Flat or Lusterless Miscellaneous (Black), Color Number 37038 of

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FED-STD-595, and Markings, Warning/Danger shall be Flat or Lusterless Red, color number 31136 of FED-STD-595. The center of gravity of the light cart shall be stenciled on the unit within 1.0 inch of the calculated center of gravity.

3.3.3 Identification and information plates.

3.3.3.1 Identification plate. An identification plate in accordance with MIL-STD-130 shall be securely attached to the light cart in a readily accessible location. The “CE” marking shall be affixed in accordance with EU requirements on or adjacent to the identification plate.The identification plate shall contain the following information:

a. Nomenclature

b. Part number

c. kW rating

d. Serial number

e. Date of manufacture

f. Manufacturer’s name

g. Manufacturer’s Commercial and Government Entity (CAGE) code

h. Contract number

i. Date of warranty expiration

j. National stock number (NSN)

3.3.3.2 Transportation data plates. A transportation data plate (s) shall be securely attached to the light cart in readily accessible location(s). The plate shall contain at least the following information:

a. Side and rear silhouette views of the light cart.

b. Horizontal and vertical location of the center of gravity of the light cart in air transportable configuration, marked on the silhouette views.

c. Shipping weight.

d. Loading cubage.

e. Overall height, width, and length.

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f. Axle loads.

g. Tie down information.

3.3.3.3 Item unique identification (IUID). The light cart and any of its components for which the Government's unit cost is more than $5,000, is serially managed, or the procuring agency determines is mission essential, shall have Unique Identification (UID), also known as Item Unique Identification (IUID), information permanently affixed on or near the respective identification plate(s), marked in accordance with MIL-STD-130. UID information shall be included as both a bar code and human readable markings.

3.3.4 Environment, safety, and occupational health (ESOH).

3.3.4.1 System safety. The design of the light cart shall not contain any system safety mishap risk categories greater than medium as defined in Table III of MIL-STD-882.

3.3.4.2 Hazardous material. The design shall minimize and control hazards associated with the inclusion or use of hazardous or toxic materials and the generation of toxic or noxious gases. The light cart shall not generate or use Class I or Class II Ozone Depleting Substances (ODS) during operation, maintenance, or disposal. The light cart shall not contain or use hexavalent chromium. For the purposes of this requirement, the Class I ODS and controlled substances identified in Chapter 4 of AFI 32-7086 shall not be used in any system, component, or process.

3.3.4.3 Component protection. All space in which work is performed during operation, service, and maintenance shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to eliminate the hazard. All wires, cables, tubes, and hoses shall be supported and protected to minimize chafing and abrasion and shall be located so as to provide adequate clearance from moving parts and high operational temperatures. Grommets shall be provided wherever wires, cables, tubes, or hoses pass through bulkheads, partitions, or structural members.

3.3.4.4 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the light cart with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve stem caps shall be made of plastic.

3.3.4.5 Noise. The design shall ensure that noise created by the light cart is compatible with the environment and minimizes exposure of personnel to noise hazards during operations and maintenance activities. Audio noise sound-pressure levels emanating from engine-diesel engine set (outside the light cart enclosure) shall not exceed 72 dB at a distance of 7 meters (23 feet) when measured at 1.2 meters (4 feet) above cement surface.

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3.3.4.6 Electrostatic discharge (ESD). The design of the light cart shall preclude equipment damage due to ESD, protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking.
3.3.5 Electromagnetic interference (EMI). The light cart shall be in accordance with the following radiated emission and susceptibility requirements of MIL-STD-461: RE102 and RS103.

3.3.6 Human systems integration. The light cart shall be designed in accordance with MIL-STD-1472 for ease of operation, inspection, and maintenance, including the use of arctic mittens.

3.3.7 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for the attachment of trim items or as provided in commercial components.

3.3.8 Welders and welding. All welders shall be certified to weld in accordance with AWS D1.1/D1.1M for structural steel, AWS D1.2/D1.2M for aluminum, AWS D1.3/D1.3M for sheet steel, and AWS D1.6/D1.6M for stainless steel, as applicable. Welding procedures and all welding on the light cart shall be in accordance with AWS D1.1/D1.1M, AWS D1.2/D1.2M, AWS D1.3/D1.3M, and AWS D1.6/D1.6M, as applicable. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings. Cast components, structural steel components with an ultimate tensile strength greater than 100,000 psi, and threaded fasteners (except for weld studs and weld nuts) shall not be welded. Heat treated components shall be in the annealed, normalized condition when welded.

3.4 Environmental conditions. The system shall be tested in accordance with MIL-HDBK-781 as tailored in the following subparagraphs:

3.4.1 Operating temperature range. The light cart shall be capable of operating in ambient temperatures ranging from -25°F to 125°F.

3.4.2 Storage temperature range. The light cart shall be capable of being stored in ambient temperatures ranging from -65°F to 165°F.

3.4.3 Precipitation.

3.4.3.1 Rain. The light cart shall be capable of storage and operation during rainfall of 5-inches per hour for three consecutive hours and 10-inches per hour for 10 consecutive minutes, with winds of up to 35 knots; and with 6-inches of rain per hour impinging on the light cart at angles from vertical to 45 degrees.

3.4.3.2 Snow. The light cart shall be capable of storage and operation during accretion of wet snow up to 2-inches per hour for at least 12 hours on exposed horizontal surfaces.

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3.4.3.3 Ice. The light cart shall be capable of storage and operation with ice accretion up to 1.5-inches on exposed horizontal surfaces. An operator may use an ice scraper for five minutes during the start-up process.

3.4.4 Solar radiation. The light cart shall not be adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments and high altitudes.

3.4.5 Fungus. All materials used in the light cart shall be fungus resistant or shall be suitably treated to resist fungus. Materials treated for fungus resistance shall retain their original electronic and physical properties, shall not present toxic hazards, and treatment shall last for the entire service life of the part. The light cart shall be suitable for operation and storage in conditions encountered in a tropical environment.

3.4.6 Salt fog. The light cart shall be capable of storage and operation in high temperature, high humidity, salt laden, sea coast environments without damage or deterioration of performance.

3.4.7 Sand and dust. The light cart shall be capable of storage and operation during exposure to wind-blown sand or dust without damage or deterioration of performance.

3.4.8 Wind resistance. Stabilization jacks shall provide sufficient support to prevent damage - with mast fully extended and equipped with four floodlights - during winds up to 60 MPH (continuous) from any direction. The light cart shall operate safely, without damage, in 70 MPH wind, with the hoist or mast lowered, with tie-down equipment installed.

3.5 Weight and dimensions. Overall weight and dimensions, driven by the air cargo transport restrictions (see 3.6.2) shall not exceed:

Weight (wet)	2500 pounds
Length (light fixtures stowed)	150 inches
Width (outriggers stowed)	67 inches
Height (mast lowered)	73 inches.

3.6 Transportability.

3.6.1 Surface transportability. The light cart shall be transportable via all modes of surface shipment (highway, rail, and water) in accordance with MIL-STD-1366, and shall be capable of withstanding the mechanical shock and vibration characteristics of highway, rail, and water transport, except that design for rail impact testing (see 5.2.5 of MIL-STD-1366) is not required.

3.6.2 Air transportability. The light cart shall be transportable on all current military cargo aircraft in accordance with MIL-STD-1791. The light cart will be configured for storage/transportation prior to movement. This includes but is not limited to securing the mast in

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the horizontal, stowed position, retracting and securing the stabilizations bars, securing the tow bar, closing and locking the cabinet doors/panels, and securing all other items/accessories that may become a danger to personnel or equipment. In all air transport configurations, the light cart shall be capable of being restrained and withstanding, without loss of serviceability, 2.0G up and 4.5G down accelerations, and shall be capable of being restrained and withstanding, without loss of structural integrity, 3.0G forward, 1.5G aft, and 1.5G lateral accelerations. The light cart shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure. The light cart shall roll on and off the aircraft, negotiating the required maximum ramp angles without shoring.

3.6.2.1 Shoring. The light cart shall be air transportable without shoring.

3.6.2.2 Axle weight. Axle weight shall not exceed 2500 pounds.

3.6.2.3 Tire pressure. Tire pressure shall not be reduced for air transport.

3.6.2.4 Winch points. Winch points shall be provided at the front and rear of the light cart for aircraft loading utilizing the aircraft's winch.

3.6.2.5 Equipment removal and reconfiguration. All equipment removed shall be stored on the light cart; caps and plugs shall permit moving and storage in transport configuration.

3.6.3 Tie downs. The light cart shall be symmetrically restrained during air and ground transport with no less than four tie downs in accordance with 5.2.1 of MIL-STD-209. Tie down points shall be rated at a minimum of 2,500 pounds, marked for capacity, with a clear opening compatible with MIL-DTL-25959 and MIL-PRF-27260 tie down devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the light cart. The tie down provisions shall be in accordance with 4.1 through 4.12 of MIL-STD-209.

3.6.4 Lifting provisions. The light cart shall incorporate fixtures permitting external lifting by crane, forklift, lifting sling or lifting eye; each attachment point shall be marked "Lift Point". The light cart shall incorporate forklift tine guides and be capable of being lifted evenly by a 10K Forklift. The lifting provisions shall be in accordance with 5.1 through 5.1.4 of MIL-STD-209.

3.7 Reliability. No reliability program is specified herein.

3.8 Maintainability. The light cart shall be designed for maintainability in accordance with 5.9 through 5.9.18 of MIL-STD-1472; forces shall not exceed those specified for both males and females.

3.8.1 Preventive maintenance. Drain plugs and filters shall be directly accessible from the ground and oriented to have unimpeded drainage to a catch pan. The light cart shall be designed with maximum usage of sealed lifetime lubrication bearings.

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3.8.2 Calibration and maintenance adjustments. The design of the light cart shall provide for readily accessible calibration adjustments and maintenance adjustments. Any calibration adjustments shall be provided by variable value components adjustable by the use of a simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration interval shall be a period of one year or greater, based on an operating time of 2,000 hours. The light cart shall be able to be serviced and visually inspected without removal of panels or components. All major components of the light cart shall be easily removable for ease of maintenance.

3.8.3 Inspection and servicing provisions.

a. Pre-use inspections and servicing tasks shall not require tools.

b. Routine service and preventive maintenance shall not require special tools (see 6.3.3).

c. Drain plugs and filters shall be directly accessible from the ground and oriented to have unimpeded drainage to a catch pan.

d. The light cart shall be designed with maximum usage of sealed lifetime lubrication bearings.

3.8.4 Relative accessibility. Critical items that require rapid maintenance shall be most accessible. When relative criticality is not a factor, items that require the most frequent access shall be most accessible. High failure rate items shall be accessible for replacement without moving non-failed items.

3.8.5 Error-proof design. The design of the light cart shall incorporate error-proofing in equipment mounting, installing, interchanging, connecting, and operating.

3.8.5.1 Mounting. Equipment shall include physical features (for example, supports, guides, size, or shape differences, fastener locations, and alignment pins) that prevent improper mounting. In the absence of physical features, equipment shall be labeled or coded to identify proper mounting and alignment.

3.8.5.2 Interchangeability. Equipment that has the same form and function shall be interchangeable throughout a system and related systems. If equipment is not interchangeable functionally, it shall not be interchangeable physically. Engine interchangeability and replaceability shall be in accordance with 3.13.6.1 through 3.13.6.3 of SAE ARP1247.

3.8.5.3 Connectors. Connectors serving the same or similar functions shall be designed to preclude mismatching or misalignment.

3.8.6 Special tools. The design of the light cart shall minimize the requirement for special tools (see 6.3.3). All special tools shall be provided with, and securely stored on, the light cart.

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3.8.7 Diagnostic software. A copy of any diagnostic software required or recommended for maintaining the light cart shall be provided with each light cart on CD ROM or DVD ROM with provisions to update diagnostic software.

3.9 Mobility. The light cart shall meet all the mobility requirements set forth in SAE AS 8090 for Type III, Group A equipment.

3.9.1 Terrain.

3.9.1.1 Paved surfaces. The light cart shall be towable at speeds up to 55 miles per hour (mph) over paved surfaces.

3.9.1.2 Graded gravel. The light cart shall be towable at speeds up to 10.0 mph over graded gravel.

3.9.1.3 Belgian block. The light cart shall be towable at speeds up to 8.0 mph over Belgian block.

3.9.2 Towing. The towing force shall be in accordance with 3.3.4 of SAE-AS8090. With the floodlight set in the stowed position and ready for towing, the light cart shall be towable over improved and unimproved road conditions. The light cart, when towed on a smooth, paved surface, shall follow the towing vehicle without wandering over 3 inches to either side of the path of the towing vehicle, and shall not induce loads of a magnitude or nature that would cause noticeable steering difficulties to the towing vehicle at towing speeds up to 55 mph. When towed on uneven surfaces (see 6.2.2.2) at speeds up to 10 mph, any resulting forces produced shall not cause any damage to the light cart.

3.9.3 Clearances. Ground clearance, ramp breakover angle, and angles of approach and departure shall be in accordance with 3.3.9 of SAE-AS8090.

3.9.4 Suspension. The light cart shall have a suspension system in accordance with 3.4.1.2 of SAE-AS8090.

3.9.5 Steering. The light cart shall have a steering system in accordance with 3.5.1.1 of SAE-AS8090.

3.9.6 Brakes.

3.9.6.1 Parking brakes. The light cart shall have parking brakes in accordance with 3.6.1 of SAE-AS8090.

3.9.6.2 Service brakes. The light cart shall have parking brakes in accordance with 3.6.2 of SAE AS8090.

3.9.7 Bumpers. The light cart shall not have bumpers.

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3.9.8 Fenders. The light cart shall have fenders in accordance with 3.9 of SAE-AS8090.

3.9.9 Mudflaps. The light cart shall not have mudflaps.

3.9.10 Pintle hook. The rear pintle hook, specified in 3.13 of SAE AS8090, shall not be provided; the light cart shall be towed singly, not in trains, as specified in 3.3.8 of SAE AS8090.

3.9.11 Towbar. The light cart shall be equipped with a towbar in accordance with 3.17.1 of SAE -AS8090. It shall incorporate a light duty, 10,000-pound lunette eye (CID A-A-52464) and accommodate any pintle-hook height from 22 inches minimum through 30 inches maximum. When light cart is not actively towed, the towbar shall be stowed upright, secured, and not hinder visibility, operation, or maintenance.

3.9.12 Tires and wheels. The light cart shall be equipped with single or dual tubeless steel belted radial tires with on/off-road type tread mounted on steel disc wheel assemblies. Tire and wheel assemblies shall be identical at all wheel positions. Where non-standard tire sizes are required, commercially standard tires sizes can be used. The commercial tire size shall be comparable to the tire size specified in SAE-AS8090 and approved by the Government before used in this application. Tires and wheels shall be in accordance with the TRA-YB (*Tire and Rim Association Year Book)* requirements for this application. Inflation pressure cold shall not exceed 35 psi for 205/75D x 15 inch tubeless, load range B, with maximum load 1480 pounds

3.10 Engine and related equipment. The light cart shall be equipped with a diesel engine. Consistent with the requirement to operate on fuels containing over 15 parts per million (ppm) sulfur, the engine shall be certified to comply with the Environmental Protection Agency (EPA) non-road diesel engine emission requirements at the time of engine manufacture. The diesel engine shall provide power for four floodlights and 120/240 VAC electrical power as an auxiliary feature. The diesel engine shall be a Type I Weather-Protected Machine with moisture resistant windings, or better, as described in NEMA-MG1, paragraphs 1.25.8.1 and 1.27.7.

3.10.1 Engine starting system.

3.10.1.1 Starter. The engine shall be equipped with a 12-volt DC electric starter.

3.10.1.2 Engine starter system energy recovery. The engine starting system shall be capable of restoring the energy expended during an engine start in less than 15 minutes of engine idle under all conditions listed in 3.10.1.2.

3.10.1.3 Engine starting aids. The engine shall start within 15 seconds cranking in any ambient temperature within the required operating range of the light cart. Internal engine starting aids, and fluid starting aids may be used prior to and during the start period to facilitate engine starting under the following conditions listed in Table 1:

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TABLE I. Engine starting aids.

Temperature Range	Starting Aids Permitted
40°F through 130°F	None
0°F through 39°F	Internal engine starting aids and fluid starting aids
-25°F through -1° F	Internal engine starting aids, fluid starting aids, and heat from the winterization system

3.10.2 Engine air intake system. The engine air intake system shall be in accordance with 3.13.1.4.3 of SAE ARP1247. The inlet shall not draw air from directly beneath the light cart and shall not be located near the cooling system air outlet nor the engine exhaust outlet. Joints shall be minimized between the air filter outlet and the actual engine air inlet and shall be designed to ensure no leakage of unfiltered air into the engine. A differential pressure air filter service indicator shall be provided.

3.10.3 Engine cooling system. The engine cooling system shall be in accordance with 3.13.1.4.2 of SAE ARP1247. Silicone radiator and heater hoses, constant-torque clamps, a coolant filter, and a coolant recovery system shall be provided. Engine coolant shall be in accordance with A-A-52624, Type I. Unless the engine manufacture specifies differently, the engine out (top of radiator) coolant temperature shall not exceed 210°F.

3.10.4 Engine lubrication system. The engine lubrication system shall be designed so that the light cart can be operated on an incline in any plane.

3.10.4.1 Engine oil. The engine shall be compatible with Grade 15W40 of MIL-PRF-2104 from 0°F to the highest temperature specified for the end item. Oil pre-heat for operation below 0°F is allowed. The engine shall be compatible with Artic engine oil per MIL-PRF-46167 from -40°F to 60°F.

3.10.4.1.1 Engine oil operating temperature. Unless the engine manufacturer specifies differently, the engine oil sump temperature shall not exceed 250°F at any operating condition.

3.10.4.1.2 Engine oil consumption. The engine oil consumption shall not exceed 0.0035 pounds per brake horsepower-hour (lbs/bhp-hr) under any operating condition.

3.10.4.2 Engine oil filter. The engine oil filter shall be in accordance with 3.13.1.4.4 of SAE-ARP1247.

3.10.5 Exhaust system. The exhaust system shall be constructed of stainless steel. The muffler(s) shall be constructed of aluminized steel or stainless steel. Exhaust system outlet(s) shall be directed away from personnel accessing any control panel or equipment compartment and the engine air intake, and shall not be directed toward the ground.

3.10.6 Engine fuels and fuel system.

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3.10.6.1 Engine operating fuels. The following shall be standard operating fuels:

a. 1-D S15, 1-D S500, 1-D S5000 in accordance with CID A-A-52557 and ASTM D975, below 60° F ambient air temperature.

b. 2-D S15, 2-D S500, 2-D S5000 in accordance with CID A-A-52557 and ASTM D975, above 32° F ambient air temperature.

c. JP-5, in accordance with MIL-DTL-5624, -40°F to 130°F ambient air temperature.

d. JP-8, in accordance with MIL-DTL-83133, -40°F to 130°F ambient air temperature.

e. TS-1 in accordance with GOST 10227-86 with JP-8 additives.

3.10.6.2 Engine emergency fuels. The following shall be emergency fuels:

Jet A, A1 in accordance with ASTM D1655, all ambient temperatures.

Note: Jet A and Jet A1 may contain corrosion inhibitor/lubricity improver and icing inhibitors which shall result in properties equal to JP-8.

3.10.6.3 Engine fuel consumption. When operating on JP-8, the engine shall meet or better the following fuel consumption at full rated load:

a) Under 20 bhp rated engine, 0.500 lbs/bhp-hr brake specific fuel consumption.

b) Over 20 bhp rated engine, 0.380 lbs/bhp-hr brake specific fuel consumption.

3.10.6.4 Fuel system. The fuel system shall be in accordance with 3.13.1.5.1 through 3.13.1.5.11 of SAE-ARP1247 except as otherwise specified herein. The fuel system shall be constructed of materials which are compatible with the fuels listed in 3.10.6.1 and 3.10.6.2. Copper shall not be used in the fuel system. The fuel system shall be equipped with a fuel shut-off valve(s) to prevent continuous spillage when fuel lines are disconnected for service.

3.10.6.4.1 Fuel priming pump. The end item shall be equipped with a 12-volt electric fuel pump in addition to the mechanical fuel pump. The electric pump shall be used as a priming pump capable of re-priming the engine fuel system following fuel exhaustion.

3.10.6.4.2 Fuel filters. Primary and secondary fuel filters and a heated fuel/water separator shall be provided. The fuel/water separator shall include a water coalescer and a drain valve that is readily accessible by an operator or a mechanic. A combination fuel filter and fuel/water separator may be provided. Fuel filter elements shall be easily replaceable by a mechanic without loss of engine prime.

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3.10.6.4.3 Fuel tank. The fuel tank shall be in accordance with 3.13.1.5.5 through 3.13.1.5.9 of SAE-ARP1247. The fuel tank shall be of sufficient capacity to permit full load operation for a minimum of 18 hours and is desired to have 20-30 gallon capacity. The fuel tank shall be removable from the light cart to permit cleaning and shall contain a fuel shut-off valve at the outlet connection. The tank shall be provided with corrosion protection and baffles. A 0.25 to 0.375-inch nominal drain valve shall be provided for emptying fuel and sediment into a container underneath the light cart without removal of the tank or any other major component. The fuel tank shall have a fuel fill opening of not less than three inch inside diameter and shall be designed to drain fuel spillage overboard for collection outside the light cart. The fuel cap shall be equipped with a retention device to prevent loss and becoming a hazard to aircraft. The fuel fill opening, fuel cap, and fuel cap retention device shall be fabricated from non-sparking material.

3.10.7 Engine diagnostic and emergency shutdown systems.

3.10.7.1 Engine diagnostic system. If the engine is equipped with an electronic control module, a diagnostic system shall be provided with a means to indicate engine faults; it shall be equipped with a CAN buss connector. If the light cart is equipped with a diagnostic or built-in-test system, the engine diagnostic system shall be integrated with it; if not, it shall be a standalone system.

3.10.7.2 Engine emergency shutdown system. The engine shall be equipped with an engine emergency shutdown system consisting of a fuel cutoff solenoid activated by the following conditions:

a. Low oil pressure, less than 10 psi or in accordance with the engine manufacturer's recommendations.

b. Coolant over heat or cylinder head temperature over heat condition in accordance with the engine manufacturer's recommendations.

c. Engine over speed shall the engine exceed 110% rated speed. This is required for engines over 100 bhp and optional for engines less than 100 bhp rated horsepower.

3.10.8 Engine operator instruments. Instrumentation and controls shall follow commercial practices. The light cart shall have an easily accessible engine-diesel engine set control panel. The control panel shall house and contain all controls, meters, switches, indicators, etc. necessary to operate the light cart. The control panel will incorporate the following equipment:

a. Hour meter (engine running time)

b. AC Voltmeter, 120V / 240V (can be switched)

c. DC Voltmeter, for battery charging system

d. Water or engine temperature gauges

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e. Oil pressure gauge

f. Start-stop/preheat/glow-plug switch

g. Fault lamp(s)

h. Panel illumination lights, circuit breakers

i. Ground Fault Circuit Interrupter (GFCI) protection for auxiliary AC power outlets (circuit breaker switched power for the lights shall not be fed from the GFCI(s)).

j. Low coolant or loss of cooling flow in the case of an air cooled engine

k. Fuel level

l. Water in fuel/water separator

3.11 Electrical system. The light cart shall be a 12-volt electrical system and have the electrical requirements listed below. Electrical properties shall be adequate to fulfill the requirements listed below and shall be the continuous duty ratings of the units.

3.11.1 Alternator. A single or dual alternator charging system in accordance with 3.13.1.4.9 of SAE-ARP1247 shall be provided.

3.11.2 Batteries and battery compartment.

3.11.2.1 Batteries. Batteries shall be of the commercial maintenance-free sealed lead acid, starved electrolyte, gas recombination, spiral wrapped, absorbent gas mat (AGM), top-post type in accordance with MIL-B-18013.

3.11.2.2 Battery compartment. The batteries shall be enclosed in a weatherproof, corrosion-resistant box or compartment and shall be readily accessible.

3.11.2.3 Battery cables. The battery cables shall be sized to handle the system voltage and current levels, be clearly identified with "+" and "-" or red and black markings, and shall not be spliced.

3.11.3 Distribution system. All electrical components shall conform to applicable UL standards (including UL486A, UL498, UL1077) and to NFPA 70. The installation of all electrical components and systems shall conform to NFPA 70. The power distribution equipment shall consist of the power distribution panel, ballasts for the metal halide lamps, switches, receptacles, interconnecting wiring, fittings, and accessories, installed and connected as required to provide a complete, operable system.

3.11.3.1 Power distribution panel. The power distribution panel shall be a circuit breaker panel mounted in a readily accessible location within the floodlight set housing. Sufficient circuit

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breakers shall be contained in the panel to provide independent control and circuit protection as defined in NFPA 70.

3.11.3.2 Circuit breakers. The circuit breakers shall conform to the requirements of UL489, and UL1077. All circuit breakers shall be approved for use as switches and shall be marked as such. Circuit breakers shall be sized to have trip ratings as required for protection against overloads, short circuits, and grounding in accordance with NFPA 70.

3.11.3.3 Wiring. Conductors for interconnecting wiring shall be composed of annealed copper wire and have a flame-retardant and oil-resistant covering conforming to NFPA 70. Extra flexible conductors shall be used where relative movement between parts may occur. Wire connectors and terminals shall be the crimp type where practicable and shall be attached in a manner to fully protect the insulation; all wire connectors shall be copper and conform to UL 486A. Conductors shall be properly clamped in a neat manner and protected by grommets or other suitable protection from abrasion where the conductors pass through metal.

3.11.3.4 Tower electrical cord. A heavy-duty, flexible, permanently coiled, retractable type power cord shall furnish power to the top of the tower mast. The supply end of the cord shall terminate in the storage housing.

3.11.3.5 Plugs and connectors. Outlet receptacles shall be of the grounding type and shall be in accordance with UL 498, NEMA-WD1 and NEMA-WD6, as applicable. All receptacles shall be weather protected and equipped with spring loaded cover plates. The light cart shall have the following GFCI and non-GFCI circuit breaker protected receptacle outlets for operating auxiliary outlets:

115 VAC/60 Hz/single phase, one (1) each:

a. 20 Amp (NEMA-5-20R) Receptacle Rated 125 VAC

b. 30 Amp (NEMA-5-30R) Receptacle Rated 125 VAC

c. 20 Amp (NEMA-5-20R) Receptacle Rated 125 VAC (GFCI protection is not required and labeled as “Non GFCI”)

230 VAC/60 Hz/single phase, one each:

a. 20 Amp (NEMA-6-20R) Receptacle Rated 250 VAC

b. 30 Amp (NEMA-L14-30R) Receptacle Rated 125/250 VAC

3.11.3.6 Ground fault protection. GFCI protection is not required for the floodlight fixtures and for one 20-amp (NEMA 5-20R) Receptacle Rated 125 VAC. Each outlet shall be marked with voltages and current rating. Each duplex receptacle shall be connected to only one circuit so that one circuit breaker protects one duplex receptacle load. The input to each pair of ballast shall be

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connected to a multi-wire 120/240V circuit with a common neutral so that two circuit breakers, one for each ungrounded conductor, protect the ballasts.

3.11.4 Lighting.

3.11.4.1 Flood lamps. Each floodlight fixture shall consist essentially of a cast, pressed, or spun aluminum, weatherproof housing with a heat-resistant clear glass lens, and weather-resistant seal. The securing devices shall be captive with either the housing or door. Provisions shall be made to release the securing devices and remove the lamp without special tools. The securing devices shall maintain the weather-resistant seal under the wide temperature changes, adverse weather, shock, and vibrational stress encountered under rough service conditions. Metal halide BT37 lamps with a rated life of at least 12,000 hours, which produce white light, shall be used. Initial light output for each 1000W lamp shall be not less than 110,000 lumens.

3.11.4.2 Ballasts. The ballasts for the four metal halide lamps shall be auto regulated (CWA) type designed to operate with 120/240V input. Each ballast shall be the enclosed, constant wattage, auto-transformer type, and shall be designed for operation at temperatures down to -25 °F (-32 °C). The ballast shall be mounted near the power distribution panel in a protected location inside the storage compartment. The line side of the ballasts shall be connected to the circuit breakers in the distribution panel.

3.11.4.3 Illumination. Lighting shall be provided for all meter faces and controls so that the light cart may be operated under any ambient lighting condition. The lighting load shall be supplied by the power source and shall be measured as part of the total electrical load. All lighting shall be controlled by the means of a single switch located on the control panel.

3.11.4.4 Compartment interior lighting. A small light shall be installed to illuminate the interior of the storage compartment. The light shall accommodate a 12V, direct current, incandescent lamp not larger than 40W.

3.12 Performance.

3.12.1 Chassis. The light cart chassis shall be of the 2 wheel type and shall meet all the requirements of SAE-AS8090 for Type III, Group A chassis except as noted in this document. The contractor may waiver sections of SAE-AS8090 that are not applicable in this application with proper justification and prior Government approval.

3.12.2 Stabilization jacks. Adjustable swivel type jacks stands shall be securely attached to extendible and retractable outriggers to level and stabilize the complete, free-standing floodlight set on a 5-degree slope. Means shall be provided for locking the jacks and outriggers in a retracted position which will not obstruct towing. When locking pins are used, they shall be of the captive type with chain or rope. Level indicators shall be provided in accessible protected locations on the trailer chassis to facilitate leveling the floodlight set prior to starting operation. The jacks shall be equipped with swivel foot pads. All jack components shall be designed and treated to eliminate corrosion. All pockets were water can enter shall be treated on internal surfaces for corrosion and have drain holes. The stabilization jacks shall provide sufficient

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support to prevent damage with full extended mast equipped with four floodlights during winds up to 60 mph continuously from any side direction.

3.12.3 Enclosure. A durable metal enclosure shall be mounted on the trailer to house all permanently mounted power distribution equipment and provide for storage of all floodlights, electrical cords, and appropriate equipment. The enclosure shall contain hinged doors to allow easy access to stored components and allow access to all components necessary to utilize the light cart. The doors shall allow easy access to components requiring servicing. The doors shall be reinforced to provide adequate rigidity and shall have fasteners to hold them securely in the closed (locked) position. Doors and hinged covers shall have a lock-open device to securely hold the door or cover in the open position, in high wind or jet blast areas, with the diesel engine set in operation. All latches and handles shall be proportioned to facilitate manipulation with heavily mitten hands and with a minimum of projection beyond the housing surface. Drip channels or similar features shall be provided over vertically hinged doors and covers. Doors and covers shall be joined with the enclosure in a manner that will meet environmental requirements and reduce noise resulting from vibration. Means shall be provided for drainage of water entrapped within the storage compartment resulting from operation with doors or covers open. These covers or doors shall not increase the overall dimensions of the light cart by more than 25 percent when open, and all corners and edges shall be sufficiently rounded to prevent injury to personnel. The top of the light cart shall be of a design to not allow the entrance or retention of foreign material (rain, snow, dust, sand, etc.).

3.12.4 Structural support. The exterior structure shall be reinforced as required to mount and support the lighting fixtures and related gear when stowed in the travel positions.

3.12.5 Lighting mast. The light tower shall be capable of being mechanically elevated and telescoped. The tower assembly shall have provisions for securely mounting four 1000W size luminaries when in operating position with sufficient spacing between fixtures to prevent contacting each other when in operation or stowed. The tower shall have four grounding and locking type receptacles mounted in one weatherproof box near the upper end of the top section of the tower. The box shall have a weatherproof cover plate. The box shall supply power to the 1000W luminaries.

3.12.5.1 Mechanical telescoping tower. The telescoping tower shall be a fabricated assembly that is commercial proven design of the 30-foot tower class. This telescoping tower, when measured in the upright (operating) position, is continuously adjustable within two feet from a fully lowered position of not greater than 15 feet to a height of not less than 30 feet above ground to the top of top most light fixture. The lower portion of the tower shall be enclosed in a manner to protect operating personnel from injury while the tower is being extended, retracted, or rotated on vertical axis up to 355 degrees. The tower assembly shall be hinged or pivoted in such a manner that, with the tower assembly in the fully retracted position, the assembly may be lowered into a horizontal (transport) position. The floodlight set shall have a positive locking mechanism to secure the tower assembly for all specified operating conditions and secure the tower assembly to the trailer-storage compartment assembly for all specified transport conditions. The floodlight set shall have a manual tower operating mechanism such that one man can raise the tower assembly from the horizontal (transport) to vertical (operating) position and

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extend the tower to the maximum height in not more than 5 minutes. The tower assembly shall have a tensioning mechanism to assure that the tower assembly is completely under control of the operating mechanism during extending and retracting operations. If a hand winch is used to raise and lower the tower the winch mechanism shall be corrosion protected with a zinc plate and the cable shall be manufactured from corrosion resistant stainless steel. The tower assembly shall have structural strength to withstand 60 mph winds from any direction, without damage, when fully extended with the luminaries mounted.

3.13 Workmanship. The light cart, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of welding, painting, and riveting; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

3.13.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.13.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3.13.3 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of excessive backlash.

3.13.4 Cleaning. The light cart shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly.

3.14 Compatibility. The light cart shall be designed and constructed in accordance with all applicable European Union (EU) requirements.

3.15 Commonality. Design shall maximize use of industry-standard commonly available components. Maximize use of components described by a commercial item description document or national stock number (NSN).

4. VERIFICATION

4.1 Classification of inspections. Requirements shall be verified in accordance with Table II, Requirement verification matrix. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

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c. Operational inspection (see 4.4).

TABLE II. Requirement verification matrix.

Section 3 Requirement	Verification Method	Section 4 Verification
3.1 First article.	Inspection	4.2 First article inspection.
3.2 Light cart description.	N/A	
3.3 Design and construction.	Inspection	4.3 Operation Inspection.
	Certification	4.6.1 Examination of product.
3.3.1 Materials, protective coatings, and finish.	N/A	
3.3.1.1 Recycled, recovered, or environmentally preferable materials.	Certification	4.6.1 Examination of product.
3.3.1.2 Protective coatings.	Certification	4.6.1 Examination of product.
3.3.1.2.1 Surface preparation and pretreatment.	Examination	4.6.1 Examination of product.
3.3.1.2.2 Primer.	Examination	4.6.1 Examination of product.
3.3.1.2.2.1 Ferrous surfaces.	Examination	4.6.1 Examination of product.
3.3.1.2.2.2 Aluminum surfaces.	Examination	4.6.1 Examination of product.
3.3.1.2.3 Topcoat.	Examination	4.6.1 Examination of product.
3.3.1.3 Dissimilar metals.	Examination	4.6.1 Examination of product.
3.3.1.4 Finish.	Examination	4.6.1 Examination of product.
3.3.1.5 Exclusion of water.	Examination	4.6.1 Examination of product.
	Test	4.6.6.3.1 Rain Test.
3.3.1.5.1 Fluid traps and faying surfaces.	Examination	4.6.1 Examination of product.
	Test	4.6.6.3.1 Rain Test.
3.3.1.5.2 Ventilation.	Examination	4.6.1 Examination of product.
3.3.1.5.3 Drainage.	Examination	4.6.1 Examination of product.
	Demonstration	4.6.2.5 Drainage Demonstration
3.3.2 Markings.	Examination	4.6.1 Examination of product.
	Test	4.6.8.1 Weight and center of gravity test.
3.3.3 Identification and information plates.	N/A	
3.3.3.1 Identification plate.	Examination	4.6.1 Examination of product.
3.3.3.2 Transportation data plate.	Examination	4.6.1 Examination of product.
3.3.3.3 Item unique identification (IUD).	Examination	4.6.1 Examination of product.

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TABLE II. Requirement verification matrix – Continued

Section 3 Requirement	Verification Method	Section 4 Verification
3.3.4. Environment, safety, and occupational health (ESOH).	N/A	
3.3.4.1 System safety.	Analysis	4.6.2 System safety hazard analysis.
3.3.4.2 Hazardous material.	Certification	4.6.1 Examination of product.
3.3.4.3 Component Protection.	Examination	4.6.1 Examination of product.
	Demonstration	4.6.10.1 Preventive maintenance demonstration..
3.3.4.4 Foreign object damage (FOD).	Examination	4.6.1 Examination of product.
3.3.4.5 Noise.	Test	4.6.3 Noise test.
	Inspection	4.3 Operational inspection.
3.3.4.6 Electrostatic discharge (ESD).	Analysis	4.6.4 Electrostatic discharge analysis.
3.3.5 Electromagnetic interference (EMI).	Test	4.6.5 Electromagnetic interference test.
3.3.6 Human systems integration.	Calculation	4.6.1 Examination of product.
	Inspection	4.3 Operational inspection.
3.3.7 Fastening devices.	Examination	4.6.1 Examination of product.
3.3.8 Welders and welding.	Examination	4.6.1 Examination of product.
3.4 Environmental conditions.	N/A	
3.4.1 Operating temperature range.	Test	4.6.6.1 High temperature storage and operation test.
	Test	4.6.6.2 Low temperature storage and operation test.
3.4.2 Storage temperature range.	Test	4.6.6.1 High temperature storage and operation test.
	Test	4.6.6.2 Low temperature storage and operation test.
3.4.3 Precipitation.	N/A	
3.4.3.1 Rain.	Test	4.6.6.3.1 Rain test.
3.4.3.2 Snow.	Analysis	4.6.6.3.2 Snow load analysis.
3.4.3.3 Ice.	Test	4.6.6.3.3 Ice accretion test.
3.4.4 Solar radiation.	Certification	4.6.1 Examination of product.
3.4.5 Fungus.	Certification	4.6.1 Examination of product.
3.4.6 Salt fog.	Test	4.6.6.4 Salt fog test.
3.4.7 Sand and dust.	Test	4.6.6.5 Sand and dust test.
3.4.8 Wind resistance	Calculation	4.6.1 Examination of product.

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TABLE II. Requirement verification matrix – Continued

Section 3 Requirement	Verification Method	Section 4 Verification
3.5 Weight and dimensions.	Test	4.6.8.1 Weight and center of gravity test.
	Test	4.6.8.2 Dimension measurement.
3.6 Transportability.	N/A	
3.6.1 Surface transportability.	Analysis	4.6.9.1.1 Surface transportability analysis.
3.6.2 Air transportability.	Analysis	4.6.9.2 Air transportability analysis.
3.6.2.1 Shoring.	Examination	4.6.1 Examination of product.
3.6.2.2 Axle weight.	Examination	4.6.1 Examination of product.
3.6.2.3 Tire pressure.	Examination	4.6.1 Examination of product.
3.6.2.4 Winch points.	Analysis	4.6.9.2 Air transportability analysis.
3.6.2.5 Equipment removal and reconfiguration.	Demonstration	4.6.9.3 Equipment removal and reconfiguration demonstration.
	Analysis	4.6.9.2 Air transportability analysis.
3.6.3 Tie downs.	Analysis	4.6.9.4.1 Tie down analysis.
	Test	4.6.9.4.2 Tie down test.
3.6.4 Lifting provisions.	Analysis	4.6.9.5.1 Lifting provision analysis.
	Test	4.6.9.5.2 Lifting provision test.
3.7 Reliability.	N/A	
3.8 Maintainability.	N/A	
3.8.1 Preventive maintenance.	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.8.2 Calibration and Maintenance Adjustments.	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.8.3 Inspection and servicing provisions.	Examination	4.6.1 Examination of product.
	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.8.4 Relative accessibility.	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.8.5 Error-proof design.	Examination	4.6.1 Examination of product.
	Certification	4.6.1 Examination of product.
3.8.5.1 Mounting.	Examination	4.6.1 Examination of product.
3.8.5.2 Interchangeability.	Examination	4.6.1 Examination of product.
3.8.5.3 Connectors.	Examination	4.6.1 Examination of product.
3.8.6 Special tools.	Examination	4.6.1 Examination of product.

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TABLE II. Requirement verification matrix – Continued

Section 3 Requirement	Verification Method	Section 4 Verification
3.8.7 Diagnostic software.	Examination	4.6.1 Examination of product.
	Inspection	4.3 Operational inspection.
3.9 Mobility.	N/A	
3.9.1 Terrain.	N/A	
3.9.1.1 Paved surfaces.	Test	4.6.11 Mobility tests.
3.9.1.2 Graded gravel.	Test	4.6.11 Mobility tests.
3.9.1.3 Belgian block.	Test	4.6.11 Mobility tests.
3.9.2 Towing.	Test	4.6.11 Mobility tests.
3.9.3 Clearances.	Test	4.6.8.2 Dimension measurement.
	Calculation	4.6.1 Examination of product.
3.9.4 Suspension.	Examination	4.6.1 Examination of product.
	Test	4.6.11 Mobility tests.
3.9.5 Steering.	Examination	4.6.1 Examination of product.
	Test	4.6.11 Mobility tests.
3.9.6 Brakes.	N/A	
3.9.6.1 Parking brakes.	Test	4.6.12.1 Parking brake test.
3.9.6.2 Service brakes.	Test	4.6.12.2 Service brake test.
3.9.7 Bumpers.	Examination	4.6.1 Examination of product.
3.9.8 Fenders.	Examination	4.6.1 Examination of product.
3.9.9 Mudflaps.	Examination	4.6.1 Examination of product.
3.9.10 Trailer tow coupler.	Examination	4.6.1 Examination of product.
3.9.11 Towbar.	Examination	4.6.1 Examination of product.
3.9.12 Tires and wheels.	Examination	4.6.1 Examination of product.
3.10 Engine and related equipment.	Inspection	4.3 Operational inspection.
3.10.1 Engine starting system.	N/A	
3.10.1.1 Starter.	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.10.1.2 Engine starter system energy recovery.	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.10.1.3 Engine starting aids.	Examination	4.6.1 Examination of product.
	Test	4.6.6.2 Low temperature storage and operation test.
3.10.2 Engine air intake system.	Examination	4.6.1 Examination of product.
	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.10.3 Engine cooling system.	Examination	4.6.1 Examination of product.
3.10.4 Engine lubrication system.	Examination	4.6.1 Examination of product.

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TABLE II. Requirement verification matrix – Continued

Section 3 Requirement	Verification Method	Section 4 Verification
3.10.4.1 Engine oil.	N/A	
3.10.4.1.1 Engine oil operating temperature.	Test	4.6.6.1 High temperature storage and operation test.
3.10.4.1.2 Engine oil consumption.	Examination	4.6.1 Examination of product.
3.10.4.2 Engine oil filter.	Examination	4.6.1 Examination of product.
3.10.5 Exhaust system.	Examination	4.6.1 Examination of product.
3.10.6 Engine fuels and fuel system.	N/A	
3.10.6.1 Engine operating fuels.	Examination	4.6.1 Examination of product.
	Test	4.6.6.1 High temperature storage and operation test.
	Test	4.6.6.2 Low temperature storage and operation test.
3.10.6.2 Engine emergency fuels.	Examination	4.6.1 Examination of product.
3.10.6.3 Engine fuel consumption.	Examination	4.6.1 Examination of product.
3.10.6.4 Fuel system.	Examination	4.6.1 Examination of product.
3.10.6.4.1 Fuel priming pump.	Examination	4.6.1 Examination of product.
3.10.6.4.2 Fuel filters.	Examination	4.6.1 Examination of product.
	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.10.6.4.3 Fuel tank.	Examination	4.6.1 Examination of product.
3.10.7 Engine diagnostic and emergency shutdown systems.	N/A	
3.10.7.1 Engine diagnostic system.	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.10.7.2 Engine emergency shutdown system.	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.10.8 Engine operator instruments.	Examination	4.6.1 Examination of product.
3.11 Electrical system.	Calculation	4.6.1 Examination of product.
3.11.1 Alternator.	Calculation	4.6.1 Examination of product.
3.11.2 Batteries and battery compartment.	N/A	4.6.1 Examination of product.
3.11.2.1 Batteries.	Examination	4.6.1 Examination of product.
3.11.2.2 Battery compartment.	Examination	4.6.1 Examination of product.
	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.11.2.3 Battery cables.	Examination	4.6.1 Examination of product.
3.11.3 Distribution system.	Calculation	4.6.1 Examination of product.

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TABLE II. Requirement verification matrix – Continued

Section 3 Requirement	Verification Method	Section 4 Verification
3.11.3.1 Power distribution panel.	Examination	4.6.1 Examination of product.
3.11.3.2 Circuit breakers.	Examination	4.6.1 Examination of product.
3.11.3.3 Wiring.	Examination	4.6.1 Examination of product.
3.11.3.4 Tower electrical cord.	Examination	4.6.1 Examination of product.
3.11.3.5 Plugs and connectors.	Examination	4.6.1 Examination of product.
3.11.3.6 Ground fault protection.	Examination	4.6.1 Examination of product.
3.11.4 Lighting.	N/A.	
3.11.4.1 Flood lamps.	Examination	4.6.1 Examination of product.
3.11.4.2 Ballasts.	Examination	4.6.1 Examination of product.
	Certification	4.6.1 Examination of product.
3.11.4.3 Illumination.	Certification	4.6.1 Examination of product.
3.11.4.4 Compartment interior lighting.	Examination	4.6.1 Examination of product.
3.12 Performance.	Inspection	4.3 Operational inspection.
3.12.1 Chassis.	Examination	4.6.1 Examination of product.
3.12.2 Stabilization jacks.	Calculation	4.6.1 Examination of product.
3.12.3 Enclosure.	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.12.4 Structural Support.	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.12.5 Lighting mast.	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.12.5.1 Mechanical telescoping tower.	Demonstration	4.6.10.1 Preventive maintenance demonstration.
3.13 Workmanship.	Examination	4.6.1 Examination of product.
3.13.1 Bolted connections.	Examination	4.6.1 Examination of product.
3.13.2 Riveted connections.	Examination	4.6.1 Examination of product.
3.13.3 Gear and lever assemblies.	Examination	4.6.1 Examination of product.
3.13.4 Cleaning.	Examination	4.6.1 Examination of product.
3.14 Compatibility.	Examination	4.6.1 Examination of product.
3.15 Commonality.	Examination	4.6.1 Examination of product.

4.2 First article inspection. The first article light carts shall be subjected to the analyses, demonstrations, examinations, and tests described in 4.6.1 through 4.6.12.2. The contractor shall provide or arrange for all test equipment and facilities. Unless otherwise approved by the procuring activity, all first article light carts shall be in the same configuration at all times and configuration changes shall not be made during the first article inspection. Except as otherwise specified, all testing in which the engine is operated shall be performed using JP-8 turbine fuel.

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4.3 Operational inspection. All operational inspections shall be satisfactorily completed and approved prior to final first article acceptance.

4.4 Conformance inspection. Each production light cart shall be subjected to the examination described in 4.6.1.

4.5 Inspection requirements.

4.5.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

4.5.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Ambient humidity.

4.5.3 Test rejection criteria. Throughout all tests specified herein, the light cart shall be closely observed for the following conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear. If excessive wear is suspected, the original equipment manufacturer’s (OEM’s) specifications or tolerances shall be utilized for making a determination.

e. Evidence of corrosion or deterioration.

f. Misalignment of components.

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g. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

h. Interference between the light cart components or between the light cart, the ground, and all required obstacles, with the exception of normal contact by the tires.

i. Evidence of undesirable mobility characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

j. Shutdown faults from:

(1) Engine cooling system.
(2) Engine lubrication system.
(3) Engine protective circuits.

4.5.4 Failure reporting, analysis, and corrective action system (FRACAS). A closed loop system shall be used to collect data, analyze, and record timely corrective action for all potential failures that occur during the first article and operational tests. The contractor's existing FRACAS shall be utilized with the minimum changes necessary to conform to this specification. The system shall cover all test samples, interfaces between test samples, test instrumentation, test facilities, test procedures, test personnel, and the handling and operating instructions. The contractor shall establish and maintain a FRACAS database; all information shall be entered into the database. Authorized procuring activity personnel shall have read-only access to the FRACAS database.

4.5.4.1 Problem and failure action. At the occurrence of a problem or potential failure that affects satisfactory operation of a test sample, entries shall be made in the appropriate data logs and the failed test sample shall be removed from test, with minimum interruption to the other test samples continuing on test.

4.5.4.1.1 Problem and failure reporting. A failure report shall be initiated at the occurrence of each problem or potential failure of the contractor hardware or software. The report shall contain the information required to permit determination of the origin and correction of the failure. The contractor’s existing failure report form may be used with minimum changes necessary to conform to the requirements of this specification. The failure report form shall include the information specified in a through c:

a. Descriptions of failure symptoms, conditions surrounding the failure, failed hardware identification, and operating time (or cycles) at the time of failure.

b. Information on each independent and dependent failure and the extent of confirmation of the failure symptoms, the identification of failure modes, and a description of all repair actions taken to return the test sample to operational readiness.

c. Information describing the results of the investigation, the analysis of all part failures, an analysis of the system design, and the corrective action taken to prevent failure

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recurrence. If no corrective action is taken, the rationale for this decision shall be recorded.

4.5.4.1.2 Identification and control of failed items. A failure tag shall be affixed to the failed part immediately upon the detection of any failure or suspected failure. The failure tag shall provide space for the failure report serial number and for other pertinent entries from the test sample failure record. All failed parts shall be marked conspicuously or tagged and controlled to ensure disposal in accordance with contract requirements. Failed parts shall not be handled in any manner which may obliterate facts which might be pertinent to the analysis. Failed parts shall be stored pending disposition by the authorized approval agency of the failure analysis.

4.5.4.1.3 Problem and failure investigations. An investigation and analysis of each reported failure shall be performed. Investigation and analysis shall be conducted to the level of hardware or software necessary to identify causes, mechanisms, and potential effects of the failure. Any applicable method (for example, test, microscopic analysis, applications study, dissection, X-ray analysis, spectrographic analysis) of investigation and analysis which may be needed to determine failure cause shall be used. When the removed part is not defective or the cause of failure is external to the part, the analysis shall be extended to include the circuit, higher hardware assembly, test procedures, and subsystem if necessary.

4.5.4.1.4 Failure verification. Reported failures shall be verified as actual failures or an acceptable explanation provided to the procuring activity for lack of failure verification. Failure verification is determined either by repeating the failure mode of the reported part or by physical or electrical evidence of failure (for example, leakage residue or damaged hardware). Lack of failure verification, by itself, is not sufficient rationale to conclude the absence of a failure.

4.5.4.1.5 Corrective action. When the cause of failure has been determined, a corrective action shall be developed to eliminate or reduce the recurrence of the failure. Repairs shall be made in accordance with normal field operating procedures and manuals. The procuring activity shall review the corrective actions at the scheduled test status review prior to implementation. In all cases the failure analysis and the resulting corrective actions shall be documented.

4.5.4.1.6 Problem and failure tracking and closeout. The closed loop failure reporting system shall include provisions for tracking problems, failures, analyses, and corrective actions. Status of corrective actions for all problems and failures shall be reviewed at scheduled test status reviews. Problem and failure closeout shall be reviewed to assure their adequacy.

4.6 Test methods.

4.6.1 Examination of product. Each light cart shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or tests listed in 4.6. A contractor-generated, Government-approved checklist (part of the test procedure) shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each light cart function shall

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be verified. Certifications and analyses shall be provided in accordance with Table III. Each production light cart shall be inspected to a Government-approved reduced version of the checklist.

TABLE III. Certifications and analyses.

Paragraph	Required Certifications and Analyses
3.3 Design and construction.	Contractor certification of baselined design with detailed attachments describing system
3.3.1.1 Recycled, recovered, or environmentally preferable materials.	Contractor certification describing all preferable materials used in product or during manufacture of product with MSDS attached.
3.3.1.2 Protective coatings.	Contractor certification of protective coatings used with salient characteristics, application process, and industry test standards the material was tested to.
3.3.4.1System safety.	Contractor system safety hazard analysis (see 4.6.2).
3.3.4.2 Hazardous material.	Contractor certification describing all hazardous materials used in product or during manufacture of product with MSDS attached and a certification that no prohibited hazardous materials were utilized.
3.3.4.6 Electrostatic discharge (ESD).	Contractor analysis of the electrostatic discharge requirement (see 4.6.4).
3.3.6 Human systems integration.	Contractor certification of how human factors affected design elements and how system meets human factors requirements.
3.4.3.2 Snow.	Contractor analysis of the snow load requirement (see 4.6.6.3.2).
3.4.4 Solar radiation.	Contractor certification that the light cart performance is not adversely affected by full time exposure to solar radiation, such as those conditions encountered in extreme desert environments.
3.4.5 Fungus.	Contractor certification that the materials used in construction of the light cart are fungus resistant or suitably treated to resist fungus along with the methods used to determine suitability of material.

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TABLE III. Certifications and analyses – Continued

Paragraph	Required Certifications and Analyses
3.4.8 Wind resistance.	Contractor calculations demonstrating item performance in environments with wind resistance. Calculate with 3.12.2.
3.6.1 Surface transportability.	Contractor surface transportability analysis (see 4.6.9.1) and certification that the light cart is transportable in accordance with MIL-STD-1366 as well as an analysis of what shipping and packing requirements are necessary to maintain system's operational and maintenance performance integrity.
3.6.2 Air transportability.	Contractor air transportability analysis (see 4.6.9.2).
3.6.2.4 Winch points.	Contractor air transportability analysis (see 4.6.9.2).
3.6.2.5 Equipment removal and configuration.	Contractor air transportability analysis (see 4.6.9.2).
3.6.3 Tie downs.	Contractor tie down provision analysis (see 4.6.9.4.1).
3.6.4 Lifting provisions.	Contractor lifting provision analysis (see 4.6.9.5).
3.8.5 Error-proof design.	Contractor certification of system design and a description and illustrations of how the design is error-proof, addressing all subparagraphs.
3.9.3 Clearances.	Contractor's calculations and illustrations with dimensions describing clearances.
3.10 Engine and related equipment., 3.10.3 Engine cooling system., 3.10.4 Engine lubrication system., 3.10.6.1 Engine operating fuels., and 3.10.6.2 Engine emergency fuels.	Engine manufacturer certification that the engine is in accordance with all applicable requirements, including exhaust emissions standards and fuels. Engine manufacturer application approval for the engine and its installation, including cooling system, lubrication system, and mounting system.

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TABLE III. Certifications and analyses – Continued

Paragraph	Required Certifications and Analyses
3.10.6.3 Engine fuel consumption.	Contractor calculation of the fuel consumption at each expected system load, for each type of fuel used for continuous operation and for night operations. Address refueling. Calculate in both gallons and liters.
3.11.1 Alternator.	Contractor calculation proving suitability and output of alternator.
3.11.2.1 Batteries.	Contractor calculation proving suitable battery type, size, power output, and drain rate.
3.11.3 Distribution system.	Contractor calculation of power loads, circuit breakers, wiring, grounding, and other considerations of distributing power from the engine for entire electrical system.
3.11.4.2 Ballasts.	Contractor certification that the ballasts were tested to operate at or below the low temperature requirement of the light cart (4.6.6.2).
3.11.4.3 Illumination.	Contractor certification of the foot candles delivered from the light cart at heights and distances compatible with industrial lighting practices and of the light illuminated within the compartment. Include certification of foot candles delivered for use of multiple light carts and their intervals.
3.12.2 Stabilization jacks.	Contractor calculation that stabilization jacks are of appropriate size and configuration to maintain operational integrity.

4.6.2 System safety hazard analysis. A system safety hazard analysis of the light cart shall be conducted in accordance with 4.3.1 through 4.3.8 of MIL-STD-882 to demonstrate compliance with the mishap risk requirement of 3.3.4.1 found in MIL-STD-882.

4.6.3 Noise test. Noise calculations and operational testing shall be performed. At a minimum, perform test procedure SAE-J1074 as described in MIL-STD-1474 and Method 661.2C of MIL-STD-705.

4.6.4 Electrostatic discharge analysis. An engineering analysis shall be performed to demonstrate compliance with the electrostatic discharge requirement of 3.3.4.6.

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4.6.5 Electromagnetic interference test. A first article light cart shall be tested in accordance with MIL-STD-461: RE 102 and RS 103 to demonstrate compliance with 3.3.5. Any engine fuel in accordance with 3.10.6.1 may be utilized.

4.6.6 Environmental testing.

4.6.6.1 High temperature storage and operation test. A first article light cart shall be tested in accordance with MIL-STD-810, Method 501.5, Procedures I and II, to demonstrate compliance with the high temperature storage and operating requirements of 3.4.1 and 3.4.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

4.6.6.2 Low temperature storage and operation test. A first article light cart shall be tested in accordance with MIL-STD-810, Method 502.5, Procedures I and II, to demonstrate compliance with the low temperature storage and operating requirements of 3.4.1 and 3.4.2, as well as the engine starting requirement of 3.10.1 and the alternator charging requirement of 3.11.1. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

4.6.6.3 Precipitation.

4.6.6.3.1 Rain test. A first article light cart shall be tested in accordance with MIL-STD-810, Method 506.5, Procedure I, to demonstrate compliance with 3.4.3.1.

4.6.6.3.2 Snow load analysis. An engineering analysis shall be performed to demonstrate compliance with the snow load requirement of 3.4.3.2, using a specific gravity of snow of 0.1 (Ref. 5.3 of MIL-STD-810).

4.6.6.3.3 Ice accretion test. A first article light cart shall be tested in accordance with MIL-STD-810, Method 521.3 with an ice thickness of 1.5-inches to demonstrate compliance with the ice accretion requirement of 3.4.3.3. The contractor shall identify those areas of the light cart where ice removal is required prior to operation.

4.6.6.4 Salt fog test. A first article light cart shall be tested in accordance with MIL-STD-810, Method 509.5, to demonstrate compliance with 3.4.6. Test duration shall be alternating 24-hour periods of salt fog exposure and drying conditions for 24-hour periods (two wet and two dry).

4.6.6.5 Sand and dust test. A first article light cart shall be tested in accordance with MIL-STD-810, Method 510.5, Procedures I (12 hours) and II (90 minutes per side), to demonstrate compliance with 3.4.7.

4.6.7 Weight and dimension tests.

4.6.7.1 Weight and center of gravity test. The weight and axle weights of a first article light cart shall be measured to demonstrate compliance with the weight requirement of 3.5 and the axle

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weight requirement of 3.6.2.2. The center of gravity shall be measured in accordance with 4.4 of TOP 2-2-800.

4.6.7.2 Dimension measurement. A first article light cart shall be measured to demonstrate compliance with the dimensional requirements of 3.5 and the clearance requirements of 3.9.3.

4.6.8 Transportability verification.

4.6.8.1 Surface transportability analysis. An engineering analysis shall be performed to demonstrate compliance with 3.6.1. The engineering analysis shall utilize the data for road transportation in accordance with MIL-STD-810, Method 514.6, Table 514.6C-II.

4.6.8.2 Air transportability analysis. An engineering analysis shall be performed to demonstrate compliance with the air transportability requirements of 3.6 and subparagraphs. The analysis shall include the tie downs and all major components and their ability to withstand the accelerations specified in 3.6.2. The analysis shall demonstrate the accessibility and structural adequacy of the winch points described in 3.6.2.4. The evaluation shall also include a dimensional analysis for the light cart while traversing the ramp and while loaded aboard the aircraft.

4.6.8.3 Equipment removal and reconfiguration demonstration. A first article light cart shall be configured for transport on cargo aircraft and then reconfigured for operation to demonstrate compliance with 3.6.2.5. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472.

4.6.8.4 Tie down provision verification.

4.6.8.4.1 Tie down provision analysis. An engineering analysis shall be performed to demonstrate compliance with the tie down provision requirements of 3.6.3.

4.6.8.4.2 Tie down provision test. A first article light cart shall be tested to demonstrate compliance with the tie down provision requirements of 3.6.3.

4.6.8.5 Lifting provision verification.

4.6.8.5.1 Lifting provision analysis. An engineering analysis shall be performed to demonstrate compliance with the lifting provision requirements of 3.6.4.

4.6.8.5.2 Lifting provision test. A first article light cart shall be tested to demonstrate compliance with the lifting provision requirements of 3.6.4.

4.6.9 Maintainability demonstration.

4.6.9.1 Preventive maintenance demonstration. All recommended preventive maintenance tasks shall be performed and the task times shall be recorded. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472. All preventive maintenance tasks

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recommended to be performed daily and at the routine maintenance intervals shall also be demonstrated by personnel wearing arctic mittens.

4.6.10 Mobility tests. A first article light cart shall be tested in accordance with Table VIII of SAE AS8090 to demonstrate compliance with the mobility requirements of 3.9,3.9.2, 3.9.4, and 3.9.5. Note that the maximum towing force is 75 pounds per ton of gross weight, in accordance with 3.3.4 of SAE AS8090, rather than 50 pounds per ton of gross weight as stated in Table VIII of SAE AS8090.

4.6.11 Brake tests.

4.6.11.1 Parking brake test. The light cart shall be tested in accordance with 4.5.7.1 of SAE AS8090 to demonstrate compliance with 3.9.6.1.

4.6.11.2 Service brake test. The light cart shall be tested in accordance with 4.5.7.2 and 4.5.8 of SAE AS8090 to demonstrate compliance with 3.9.6.2.

5. PACKAGING

5.1 For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The light cart is designed to be a self-contained, mobile, power generating light source for illumination of work sites.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. If first article inspection is required, specify the number of first article units (see 3.1).

c. Finish color required (see 3.3.1.40.

d. Packaging requirements (see 5.1).

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6.3 Definitions.

6.3.1 Commercial item.

(1) Any item, other than real property, that is of a type customarily used by the general public or by non-Governmental entities for purposes other than Governmental purposes, and—

(i) Has been sold, leased, or licensed to the general public; or

(ii) Has been offered for sale, lease, or license to the general public;

(2) Any item that evolved from an item described in paragraph (1) of this definition through advances in technology or performance and that is not yet available in the commercial marketplace, but will be available in the commercial marketplace in time to satisfy the delivery requirements under a Government solicitation;

(3) Any item that would satisfy a criterion expressed in paragraphs (1) or (2) of this definition, but for—

(i) Modifications of a type customarily available in the commercial marketplace; or

(ii) Minor modifications of a type not customarily available in the commercial marketplace made to meet Federal Government requirements. Minor modifications mean modifications that do not significantly alter the non-Governmental function or essential physical characteristics of an item or component, or change the purpose of a process. Factors to be considered in determining whether a modification is minor include the value and size of the modification and the comparative value and size of the final product. Dollar values and percentages may be used as guideposts, but are not conclusive evidence that a modification is minor;

(4) Any combination of items meeting the requirements of paragraphs (1), (2), (3), or (5) of this definition that are of a type customarily combined and sold in combination to the general public;

(5) Installation services, maintenance services, repair services, training services, and other services if—

(i) Such services are procured for support of an item referred to in paragraph (1), (2), (3), or (4) of this definition, regardless of whether such services are provided by the same source or at the same time as the item; and

(ii) The source of such services provides similar services contemporaneously to the general public under terms and conditions similar to those offered to the Federal Government;

(6) Services of a type offered and sold competitively in substantial quantities in the commercial marketplace based on established catalog or market prices for specific tasks performed or

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specific outcomes to be achieved and under standard commercial terms and conditions. For purposes of these services—

(i) “Catalog price” means a price included in a catalog, price list, schedule, or other form that is regularly maintained by the manufacturer or vendor, is either published or otherwise available for inspection by customers, and states prices at which sales are currently, or were last, made to a significant number of buyers constituting the general public; and

(ii) “Market prices” means current prices that are established in the course of ordinary trade between buyers and sellers free to bargain and that can be substantiated through competition or from sources independent of the offerors.

(7) Any item, combination of items, or service referred to in paragraphs (1) through (6) of this definition, notwithstanding the fact that the item, combination of items, or service is transferred between or among separate divisions, subsidiaries, or affiliates of a contractor; or

(8) A non-developmental item, if the procuring agency determines the item was developed exclusively at private expense and sold in substantial quantities, on a competitive basis, to multiple State and local Government. (Reference the Federal Acquisition Regulation (FAR) 2.101)

6.3.2 Common hand tool. A non-powered tool that is likely to be found in a typical mechanic’s toolbox. Common hand tools include open end, boxed end, combination, socket (both 6- and 12-point in both standard and deep-well), and hex key wrenches, in sizes up to and including 1-inch and metric sizes up to and including 25-mm; ratchet handles, extensions, and swivels; slotted and Phillips-head screwdrivers; regular and snap-ring pliers; and a ball-peen hammer.

6.3.3 Special tool. A tool that is not commercially and readily available from a source other than the light cart contractor.

6.4 Subject term (key word) listing.

Harsh environment
Outside usage
Self-contained
Towable

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